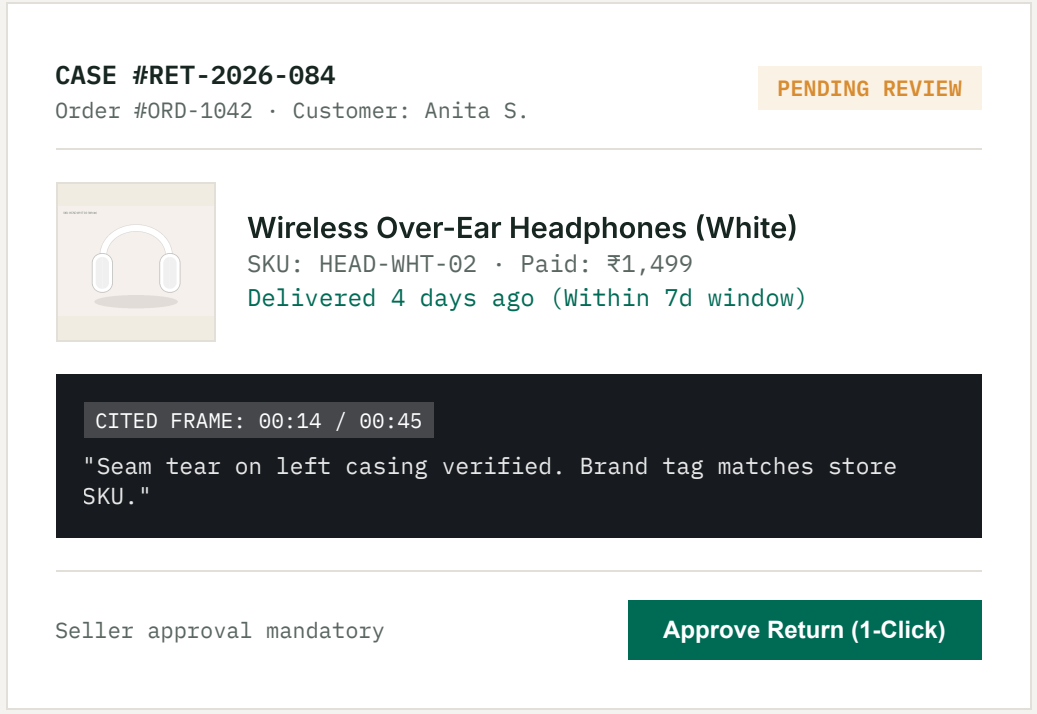
ReturnLens

Review returns with cited video evidence.

A focused return triage agent for solo direct-to-consumer sellers. It links buyer unboxing clips to store records and policy rules for 1-click decision making.

Team Rookie | 3 Members | 24-Hour Prototype | Mock Data & Staged Evidence



THE PROBLEM · SOLO SELLER BOTTLENECK

Return evidence is scattered across four sources

Sellers make high-stakes return decisions from memory while toggling between fragmented tabs.

1. BUYER CHAT / VIDEO

"Received damaged unit. Need full refund."

unboxing_clip_32s.mp4

2. ORDER RECORD

ORD-8492 · ₹1,499
SKU: HEAD-WHT-02
Customer: Rohit K.

3. DELIVERY STATUS

Delivered: 26 Oct (4d ago)
Tracking: TRK-98214-DEL
Carrier: MockPost

4. RETURN POLICY

7-day limit from delivery.
Tags intact. Original packaging.
Refund cap: ₹4,999.

The Cost of Guesswork

- **Wrong Approval:** Unrecoverable cash, shipping fees, and inventory lost to fraud or abuse.
- **Unfair Denial:** Permanent loss of customer goodwill, chargebacks, and marketplace penalties.

Sourced Industry Benchmark

Fraud estimates differ by definition: **9%** (NRF 2025, fraud only) vs **15.14%** (Appriss/Deloitte 2024, fraud & abuse).

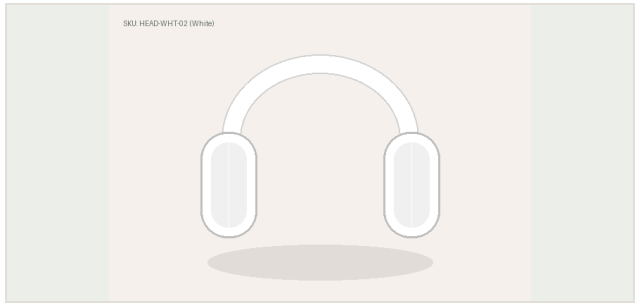
*U.S. retail estimates; not measured on our target sellers.

PRODUCT DEMONSTRATION · VISUAL EVIDENCE BENCHMARK

A color mismatch needs seller review

When video evidence contradicts store truth, ReturnLens surfaces the discrepancy without auto-denying.

STORE API RECORD — ORDER TRUTH



SKU: HEAD-WHT-02
Product: Wireless Headset
Ordered Color: White
Price: ₹1,499 · Delivered 4d ago

VIDEO EVIDENCE — STAGED FRAME [00:07]



Observed color: Black
SKU: Not verified (unreadable)
Tag: Tag not visible in this frame
Discrepancy: Color divergence detected

Policy P-03 Check: FAIL (MISMATCH)

STATUS: NEEDS SELLER REVIEW

No automatic denial. No return label or refund draft issued at this stage.

Seller Resolution Options:

- Review possible warehouse mis-shipment
- Request outside-box shipping label photo
- Hold for physical inspection upon return

Illustrative prototype · Staged mock evidence

Evidence checks and seller approval

A deterministic pipeline where models observe, code verifies, and the seller authorizes.

01 · INGEST

Receive & Retrieve

Buyer video & message via mock chat; queries order and courier APIs.

02 · OBSERVE

Model Observation

Models inspect frames and generate structured observations with timestamps.

03 · CHECK

Policy Engine (Code)

Pure code evaluates 7-day window, delivery date, SKU match, and refund cap.

04 · VERIFY

Independent Critic

Audits cited frames to confirm observations are supported by actual visual data.

05 · AUTHORIZE

Seller Gate

Seller 1-click approval permits mock return label & refund draft.

UNCLEAR EVIDENCE

Asks for specific photo (e.g. tag/serial) and resumes the exact same case.

MISMATCH / CONTRADICTION

Escalates directly to seller review with proof pack; never auto-denies.

ALL POLICY CHECKS PASS

Prepares 1-click approval package for seller; draft in `pending_return_receipt`.

AUTHORITY MODEL: Server-enforced limits · Models observe; pure code evaluates policy · Seller approval required for all actions

DEMO SCENARIOS · BENCHMARK MATRIX

Four return scenarios

Expected prototype behavior across diverse real-world return situations.

SCENARIO	CASE IDENTIFIER	EVIDENCE SITUATION	EXPECTED PROTOTYPE BEHAVIOR
Ready	Case A · ORD-1042 ₹1,499	Seam tear verified at 00:14; inside 7-day window; brand tag matches.	**Prepare for seller approval** → Generates mock label & refund draft in `pending_return_receipt`.
Mismatch	Case B · ORD-8492 ₹1,499	White ordered vs Black observed in frame 00:07; tag missing.	**Needs seller review** → Flagged with proof pack for manual decision; no auto-denial, no label created.
Unclear	Case C · ORD-5120 ₹850	4-second dark video (brightness 28); only 2 usable frames.	**Request specific evidence** → Prompts buyer for well-lit tag photo; preserves active case session.
Context	Case D · ORD-3391 ₹2,199	Continuous take tracking parcel from sealed label to unboxing.	**Check sequence continuity** → Verifies chronological context; asks for missing angles if needed.

*Expected prototype behavior on staged benchmark cases; not a measurement of live production metrics.

DELIVERY PLAN · TEAM ROOKIE

A focused 24-hour prototype

Realistic scope and role allocation for a 3-person sprint during the hackathon.

00h – 06h	**Mock APIs & Core Flow**	FastAPI mock store and shipping services, SQLite seed data, core StateGraph.
06h – 12h	**Evidence Pipeline & Policy Checks**	ffmpeg CPU frame extraction (~1 fps), multimodal prompt grounding, deterministic code policy rules.
12h – 18h	**Integration & 4 Demo Scenarios**	End-to-end testing across Cases A, B, C, D; critic checks; fallback error handling.
18h – 24h	**Demo Polish & Packaging**	Seller review console polish, 3-minute video walk-through, GitHub README, submission package.

Person A (Backend & Logic)
Agent flow, state graph, deterministic policy engine, critic, audit log.

Person B (Multimodal & UI)
Video pipeline, frame indexer, seller console, evidence pack view.

Person C (QA & Integration)
Integration, mock store data, 4 demo cases, fallback testing, demo video.

Strict 24-Hour Scope Guardrails:
Mock services · Staged evidence · Single seller return workflow · No production payments.